

SIMATS ENGINEERING
Department of Computer Science and Engineering
ASSESSMENT TOOL 3 – CODING CHALLENGE QUESTIONS

Course Code:	CSA12	Course Title:	Computer Architecture
CO Assessed:	CO1 – Precision (BL4)	Assessment:	Coding challenge questions
Weightage:	40%	Total Marks:	100
CO1 Statement:	Analyse the functional units of a computer system including CPU, memory, bus structures, and I/O components by examining instruction execution cycles, addressing modes, and performance metrics such as CPI, clock cycles, and benchmarking		
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Section / Batch:	CSE-AI	Date:	28-07-2026

Instructions to Students

- Analyze the given scenario and identify the problem requirements before developing the solution.
- Design the algorithm or flowchart and select the appropriate 8085 instructions, registers, and addressing modes required for implementation.
- Develop and execute the 8085 Assembly Language Program (ALP) using a simulator or trainer kit to solve the given problem.
- Verify and validate the output by testing the program with the given input data and ensuring the expected results are obtained.
- Interpret the program execution by explaining the logic used, the role of registers, memory locations, flags, and the final output generated.

QUESTIONS

1. A hospital stores the body temperature readings of 12 patients in consecutive memory locations. A patient is considered to have a fever if the temperature value is greater than the fever threshold stored in memory. Develop an 8085 ALP to:
 - Count the number of patients with fever.
 - Find the highest recorded temperature.
 - Store both results in memory.
2. An automated irrigation system stores water levels measured every hour. If the water level falls below the minimum safe level, an alert should be generated. Write an 8085 ALP to:
 - Count the number of low-water-level readings.
 - Determine the lowest water level recorded.
 - Store an alert flag if more than three readings are below the safe level.
3. A library stores the number of books borrowed from ten different sections. The librarian wants to identify the most frequently used section and determine the total number of books borrowed. Write an 8085 ALP to:

- Calculate the total books borrowed.
- Find the section with the highest borrowing count.
- Store both results.

Code of Conduct

I M.KOUSHIK REDDY /192372185 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _M.KOUSHIK REDDY

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Understanding	20%	Clearly understands problem constraints, edge cases, and competitive requirements	Understands problem with minor gaps in constraints	Partial understanding; misses key constraints	Misinterprets problem or constraints
Algorithm	25%	Selects optimal algorithm with best time and space complexity for competitive constraints	Uses efficient algorithm with minor scope for improvement	Uses basic/inefficient approach	No clear algorithmic approach
Code Implementation	20%	Writes correct, efficient, and clean code that passes all constraints	Code works with minor errors or inefficiencies	Partially correct code; fails for some cases	Code does not execute or gives incorrect results
Competitive Performance	20%	Solves problem within time limits and handles large inputs effectively	Solves within acceptable time with minor delays	Struggles with time limits or larger inputs	Unable to solve within time constraints

Test Case Handling	15%	Handles all test cases including edge and hidden cases	Handles most test cases	Misses important edge cases	Fails on multiple test cases
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Marks Evaluation:

Q. No	Problem Understanding (20)	Algorithm (25)	Code Implementation (20)	Competitive Performance (20)	Test case handling (15)	Total (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 20	/ 25	/ 20	/ 20	/ 15	/ 100

Course Coordinator Faculty Incharge:-

ANSWERS AND OUTPUTS

1. Hospital Temperature Monitoring

Sol:-

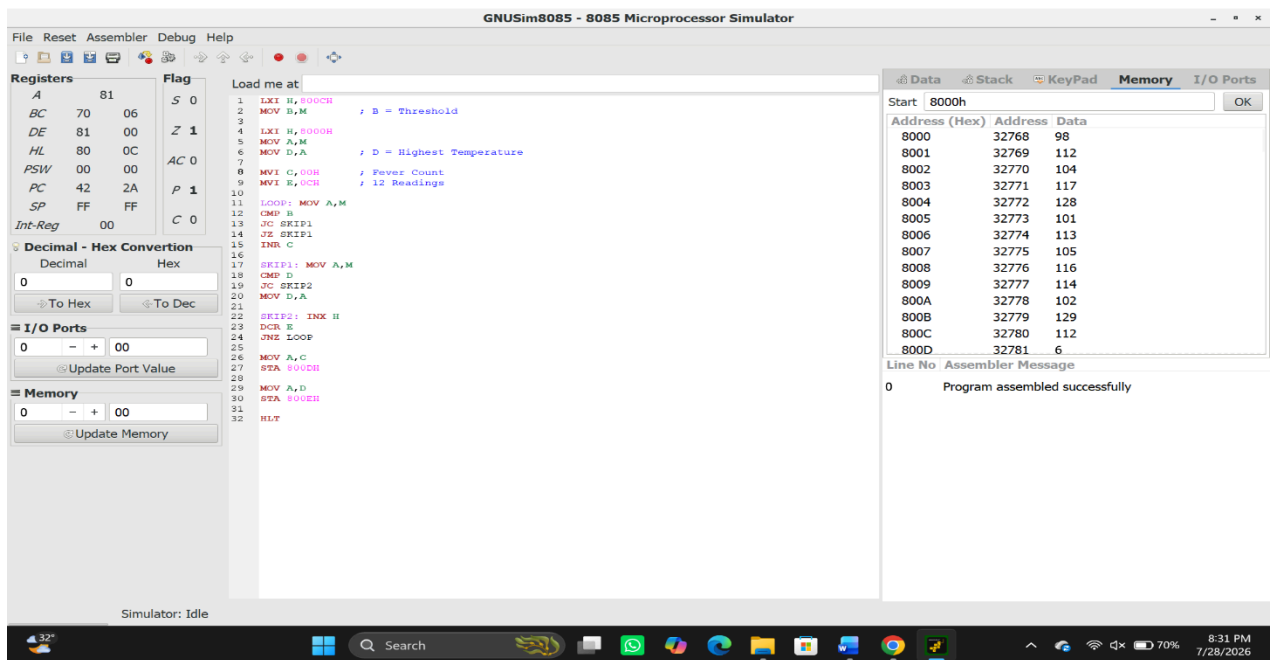
□ Aim:-

- 12 temperature readings are stored from 8000H–800BH.
- Fever threshold is stored at 800CH.
- Count patients whose temperature is greater than threshold.
- Find the highest temperature.
- Store:
 - Fever count → 800DH
 - Highest temperature → 800EH

□ Algorithm:-

1. Load fever threshold.
2. Initialize count = 0.
3. Assume first temperature is the highest.
4. Compare each of the 12 readings with threshold.
5. If temperature > threshold, increment count.
6. Compare with highest temperature and update if necessary.
7. Repeat for all 12 readings.
8. Store count and highest temperature.

□ Output and Screenshot:-



2. Automated Irrigation System

Sol:-

□ Aim:-

- Water levels stored at **9000H–900BH** (12 readings).
- Minimum safe level stored at **900CH**.
- Count readings **below safe level**.
- Find **lowest water level**.
- If count > 3, store alert flag = **01H**, otherwise **00H**.
- Store:
 - Low-water count → **900DH**
 - Lowest level → **900EH**
 - Alert flag → **900FH**

□ Algorithm:-

1. Load minimum safe level.
2. Initialize count = 0.
3. Assume first reading is the lowest.
4. Compare each reading with safe level.
5. If reading < safe level, increment count.
6. Compare with lowest value and update if needed.
7. After processing all readings:
 - If count > 3 → Alert = 01H.
 - Else Alert = 00H.
8. Store results.

□ Output and screenshot:-

The screenshot displays the GNUSim8085 - 8085 Microprocessor Simulator interface. The main window shows the assembly code for the irrigation system. The registers panel on the left shows the status of various registers, including A, BC, DE, HL, PSW, PC, SP, and Int-Reg. The memory panel on the right shows the contents of memory locations from 9000h to 9016h. The assembly code is as follows:

```
1 LXI H,900CH ; Safe Level
2 MOV B,M
3 LXI H,9000H
4 MOV A,M
5 MOV D,A ; Lowest Water Level
6
7 MVI C,00H ; Low Count
8 MVI E,0CH ; 12 Readings
9
10 LOOP: MOV A,M
11 CMP B
12 JNC SKIP1
13 INR C
14 SKIP1: MOV A,M
15 CMP D
16 JNC SKIP2
17 MOV D,A
18 SKIP2: INX H
19 DCR E
20 JNZ LOOP
21 MOV A,C
22 STA 900DH
23 MOV A,D
24 STA 900EH
25 MOV A,C
26 CPI 04H
27 JC NOALERT
28 MVI A,01H
29 STA 900FH
30 JUMP END
31 NOALERT: MVI A,00H
32 STA 900FH
33 HLT
```

The memory panel shows the following data:

Address (Hex)	Address	Data
9009	36873	50
900A	36874	46
900B	36875	57
900C	36876	50
900D	36877	5
900E	36878	43
900F	36879	1
9010	36880	0
9011	36881	0
9012	36882	0
9013	36883	0
9014	36884	0
9015	36885	0
9016	36886	0

The assembly message panel at the bottom shows the message: "Program assembled successfully".

3. 8085 ALP – Library Book Borrowing

Sol:-

□ **Aim:-**

- Number of books borrowed from **10 sections** is stored in memory.
- Calculate the **total number of books borrowed**.
- Find the **highest borrowing count**.
- Store both results in memory.

□ **Memory Locations**

Address	Purpose
8000H–8009H	10 section borrowing counts (Input)
800AH	Total books borrowed (Output)
800BH	Highest borrowing count (Output)

□ **Algorithm:-**

1. Load the first section count.
2. Assume it is the highest.
3. Initialize total with the first value.
4. Read the remaining 9 section counts.
5. Add each value to the total.
6. Compare each value with the highest count.
7. Update the highest value if required.
8. Store the total and highest count.
9. Stop.

□ **Output**

Total Books Borrowed :- $8 + 5 + 10 + 7 + 6 + 9 + 4 + 3 + 8 + 5 = 65 = 41H$

Total Books:- 10

Output and screenshot:-

The screenshot displays the GNUSim8085 - 8085 Microprocessor Simulator interface. The main window is divided into several sections:

- Registers:** A table showing the current values of the 8085 registers. The PC (Program Counter) is 42, and the SP (Stack Pointer) is FF.
- Flags:** A table showing the status of the flags. The S (Sign) flag is 0, Z (Zero) is 1, AC (Auxiliary Carry) is 0, P (Parity) is 1, and C (Carry) is 1.
- Assembly Code:** A list of assembly instructions with line numbers. The code is as follows:

```
1 LXI H,8000H
2
3 MOV A,M
4 MOV D,A ; Highest count
5 MOV C,A ; Total
6
7 MVI B,09H ; Remaining 9 values
8
9 LOOP: INX H
10 MOV A,M
11
12 ADD C
13 MOV C,A
14
15 MOV A,M
16 CMP D
17 JC SKIP
18 MOV D,A
19
20 SKIP: DCR B
21 JNZ LOOP
22
23 MOV A,C
24 STA 800AH
25
26 MOV A,D
27 STA 800BH
28
29 HLT
```
- Memory:** A table showing the memory contents. The address 8000h is selected, and the data is 8. The memory contents are as follows:

Address (Hex)	Address	Data
8000	32768	8
8001	32769	5
8002	32770	10
8003	32771	7
8004	32772	6
8005	32773	9
8006	32774	4
8007	32775	3
8008	32776	8
8009	32777	5
800A	32778	65
800B	32779	10
800C	32780	0
800D	32781	0
- I/O Ports:** A section for interacting with the I/O ports. The current value is 0, and the port address is 00.
- Message Log:** A section showing the output of the simulator. The message "Program assembled successfully" is displayed.